NW-1-2026

Data Handling

CO2-Assignment-2

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(1) Visualizing Amounts

(a) Horizontal bar chart of vehicle classes

```
library (ggplot2)
library (dplyr)
mpg %>% count (class) %>% %.
ggplot (aes (x = reorder (class, n), y = n))
+ geom_col (fill = "steelblue") +
  coord_flip () +
  labs (title = "Vehicle Counts by Class",
    x = "Class", Y = "Count")
```

Interpretation : The chart displays vehicle counts by class, arranged from highest to lowest frequency

(b) Grouped bar chart by drive type

Interpretation : The chart compares vehicle classes based on their drive types (4, f and r).

(c) Stacked bar chart of diamonds

Interpretation : The stacked chart shows diamond counts for each cut category, divided by clarity grades.

(2) Visualizing a Single Distribution

(a) Histograms with three bin widths -(2)

```
library (ggplot2)
library (patchwork)

p1 <- ggplot
p2 <- ggplot
p3 <- ggplot
  geom_histogram (binwidth = 1, fill = "red")

p1 / p2 / p3
```

Interpretation : Small bin widths show more details while large widths give a smoother and simpler distribution.

(b) Density plots with three bandwidths

Interpretation : Small bandwidth shows more peaks and details. large bandwith produces a smoother.

(3) Visualizing Multiple Distributions

(a) Overlapping density plots of iris

```
ggplot (iris, aes (Sepal.Length,
  color = Species, fill = Species)) +
  geom_density (alpha = 0.3) +
  labs (title = "Sepal Length
    Distribution")
```

Interpretation : The density plot compares sepal length distributions among three iris species using transparency.

(b) boxplots of Tooth Growth

```
ggplot(ToothGrowth, aes(factor(dose),
len, fill = supp)) +
  geom_boxplot() +
  geom_jitter(width = 0.1)
```

Interpretation : The boxplots compares tooth growth for different doses and supplement types. Individual observations.

(4) Visualizing Proportions

(a) Pie chart of drive type

```
mpg %>% count(drv) %>%
ggplot(aes(x = "", y = n, fill = drv)) +
  geom_col() +
  coord_polar("y") +
  geom_text(aes(label = paste0(round(n/
  sum(n) * 100), "%")),
  position = position_stack(vjust = 0.5))
```

Interpretation : The pie chart displays the percentage share of vehicles for each drive type.

(b) 100%. stacked bar chart

Interpretation : The chart compares the proportion of clarity grades within each diamond category.

(5) (4) Nested Proportions and X-Y Relationships

(a) Treemap of vehicle class and manufacturer

```
library (treemapify)
mpg %>% count (class, manufacturer) %>%
ggplot (aes (area = n, fill = class,
            subgroup = class,
            label = manufacturer)) +
  geom_treemap() +
  geom_treemap_text()
```

Interpretation : The treemap displays vehicle manufacturers grouped by vehicle class. Rectangle sizes represent vehicle counts.

(b) Scatterplot with regression lines

Interpretation : The scatterplot shows the relationship b/n engine displacement and highway mileage for different drive types. Regression lines show the trend for each group.

Assignment - 02-2

output

